

13 Analog-to-digital converter (ADC)

This section applies to the whole STM32F4xx family, unless otherwise specified.

13.1 ADC introduction

The 12-bit ADC is a successive approximation analog-to-digital converter. It has up to 19 multiplexed channels allowing it to measure signals from 16 external sources, two internal sources, and the V_{BAT} channel. The A/D conversion of the channels can be performed in single, continuous, scan or discontinuous mode. The result of the ADC is stored into a left- or right-aligned 16-bit data register.

The analog watchdog feature allows the application to detect if the input voltage goes beyond the user-defined, higher or lower thresholds.

13.2 ADC main features

- 12-bit, 10-bit, 8-bit or 6-bit configurable resolution
- Interrupt generation at the end of conversion, end of injected conversion, and in case of analog watchdog or overrun events
- Single and continuous conversion modes
- Scan mode for automatic conversion of channel 0 to channel 'n'
- Data alignment with in-built data coherency
- Channel-wise programmable sampling time
- External trigger option with configurable polarity for both regular and injected conversions
- Discontinuous mode
- Dual/Triple mode (on devices with 2 ADCs or more)
- Configurable DMA data storage in Dual/Triple ADC mode
- Configurable delay between conversions in Dual/Triple interleaved mode
- ADC conversion type (refer to the datasheets)
- ADC supply requirements: 2.4 V to 3.6 V at full speed and down to 1.8 V at slower speed
- ADC input range: $V_{REF-} \leq V_{IN} \leq V_{REF+}$
- DMA request generation during regular channel conversion

[Figure 44](#) shows the block diagram of the ADC.

Note: V_{REF-} , if available (depending on package), must be tied to V_{SSA} .

13.3 ADC functional description

Figure 44 shows a single ADC block diagram and Table 65 gives the ADC pin description.

Figure 44. Single ADC block diagram

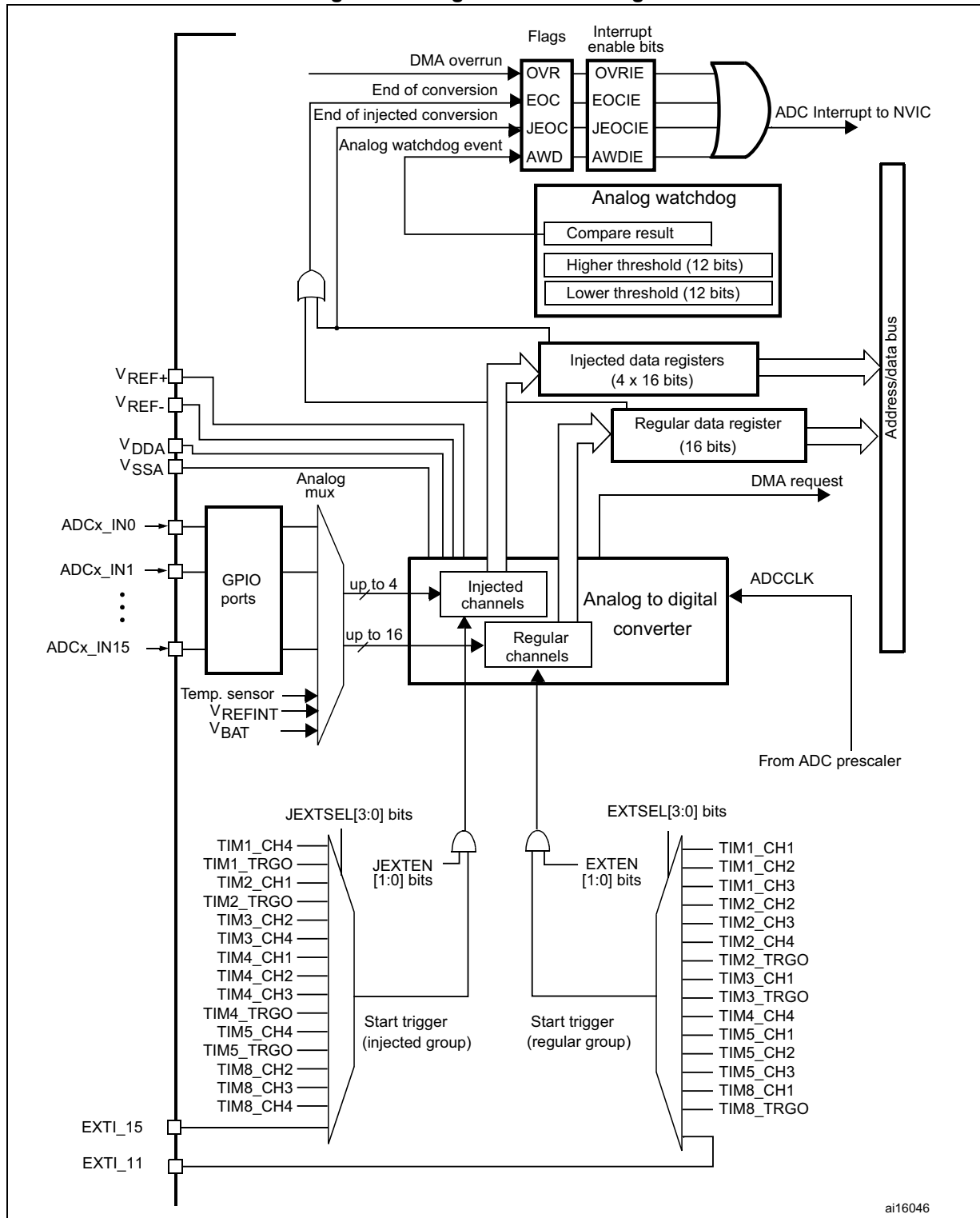


Table 65. ADC pins

Name	Signal type	Remarks
V_{REF+}	Input, analog reference positive	The higher/positive reference voltage for the ADC, $1.8\text{ V} \leq V_{REF+} \leq V_{DDA}$
V_{DDA}	Input, analog supply	Analog power supply equal to V_{DD} and $2.4\text{ V} \leq V_{DDA} \leq V_{DD}$ (3.6 V) for full speed $1.8\text{ V} \leq V_{DDA} \leq V_{DD}$ (3.6 V) for reduced speed
V_{REF-}	Input, analog reference negative	The lower/negative reference voltage for the ADC, $V_{REF-} = V_{SSA}$
V_{SSA}	Input, analog supply ground	Ground for analog power supply equal to V_{SS}
ADCx_IN[15:0]	Analog input signals	16 analog input channels

13.3.1 ADC on-off control

The ADC is powered on by setting the ADON bit in the ADC_CR2 register. When the ADON bit is set for the first time, it wakes up the ADC from the Power-down mode.

Conversion starts when either the SWSTART or the JSWSTART bit is set.

You can stop conversion and put the ADC in power down mode by clearing the ADON bit. In this mode the ADC consumes almost no power (only a few μA).

13.3.2 ADC clock

The ADC features two clock schemes:

- Clock for the analog circuitry: ADCCLK, common to all ADCs
This clock is generated from the APB2 clock divided by a programmable prescaler that allows the ADC to work at $f_{PCLK2}/2$, $/4$, $/6$ or $/8$. Refer to the datasheets for the maximum value of ADCCLK.
- Clock for the digital interface (used for registers read/write access)
This clock is equal to the APB2 clock. The digital interface clock can be enabled/disabled individually for each ADC through the RCC APB2 peripheral clock enable register (RCC_APB2ENR).

13.3.3 Channel selection

There are 16 multiplexed channels. It is possible to organize the conversions in two groups: regular and injected. A group consists of a sequence of conversions that can be done on any channel and in any order. For instance, it is possible to implement the conversion sequence in the following order: ADC_IN3, ADC_IN8, ADC_IN2, ADC_IN2, ADC_IN0, ADC_IN2, ADC_IN2, ADC_IN15.

- A **regular group** is composed of up to 16 conversions. The regular channels and their order in the conversion sequence must be selected in the ADC_SQRx registers. The total number of conversions in the regular group must be written in the L[3:0] bits in the ADC_SQR1 register.
- An **injected group** is composed of up to 4 conversions. The injected channels and their order in the conversion sequence must be selected in the ADC_JSQR register.

The total number of conversions in the injected group must be written in the L[1:0] bits in the ADC_JSQR register.

If the ADC_SQRx or ADC_JSQR registers are modified during a conversion, the current conversion is reset and a new start pulse is sent to the ADC to convert the newly chosen group.

Temperature sensor, V_{REFINT} and V_{BAT} internal channels

- For the STM32F40x and STM32F41x devices, the temperature sensor is internally connected to channel ADC1_IN16.
The internal reference voltage VREFINT is connected to ADC1_IN17.
- For the STM32F42x and STM32F43x devices, the temperature sensor is internally connected to ADC1_IN18 channel which is shared with VBAT. Only one conversion, temperature sensor or VBAT, must be selected at a time. When the temperature sensor and VBAT conversion are set simultaneously, only the VBAT conversion is performed.
The internal reference voltage VREFINT is connected to ADC1_IN17.

The V_{BAT} channel is connected to ADC1_IN18 channel. It can also be converted as an injected or regular channel.

Note: The temperature sensor, V_{REFINT} and the V_{BAT} channel are available only on the master ADC1 peripheral.

13.3.4 Single conversion mode

In Single conversion mode the ADC does one conversion. This mode is started with the CONT bit at 0 by either:

- setting the SWSTART bit in the ADC_CR2 register (for a regular channel only)
- setting the JSWSTART bit (for an injected channel)
- external trigger (for a regular or injected channel)

Once the conversion of the selected channel is complete:

- If a regular channel was converted:
 - The converted data are stored into the 16-bit ADC_DR register
 - The EOC (end of conversion) flag is set
 - An interrupt is generated if the EOCIE bit is set
- If an injected channel was converted:
 - The converted data are stored into the 16-bit ADC_JDR1 register
 - The JEOC (end of conversion injected) flag is set
 - An interrupt is generated if the JEOCIE bit is set

Then the ADC stops.

13.3.5 Continuous conversion mode

In continuous conversion mode, the ADC starts a new conversion as soon as it finishes one. This mode is started with the CONT bit at 1 either by external trigger or by setting the SWSTART bit in the ADC_CR2 register (for regular channels only).

After each conversion:

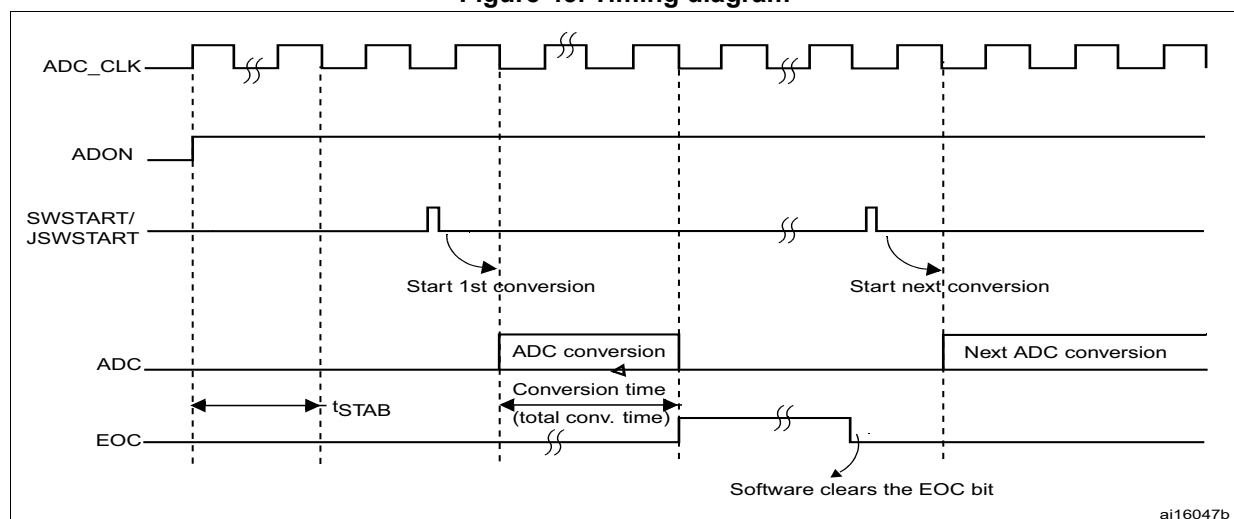
- If a regular group of channels was converted:
 - The last converted data are stored into the 16-bit ADC_DR register
 - The EOC (end of conversion) flag is set
 - An interrupt is generated if the EOCIE bit is set

Note: *Injected channels cannot be converted continuously. The only exception is when an injected channel is configured to be converted automatically after regular channels in continuous mode (using JAUTO bit), refer to [Auto-injection](#) section).*

13.3.6 Timing diagram

As shown in [Figure 45](#), the ADC needs a stabilization time of t_{STAB} before it starts converting accurately. After the start of the ADC conversion and after 15 clock cycles, the EOC flag is set and the 16-bit ADC data register contains the result of the conversion.

Figure 45. Timing diagram



13.3.7 Analog watchdog

The AWD analog watchdog status bit is set if the analog voltage converted by the ADC is below a lower threshold or above a higher threshold. These thresholds are programmed in the 12 least significant bits of the ADC_HTR and ADC_LTR 16-bit registers. An interrupt can be enabled by using the AWDIE bit in the ADC_CR1 register.

The threshold value is independent of the alignment selected by the ALIGN bit in the ADC_CR2 register. The analog voltage is compared to the lower and higher thresholds before alignment.

[Table 66](#) shows how the ADC_CR1 register should be configured to enable the analog watchdog on one or more channels.

Figure 46. Analog watchdog's guarded area

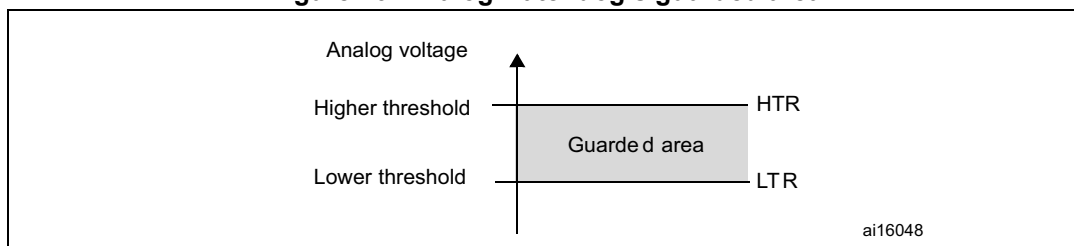


Table 66. Analog watchdog channel selection

Channels guarded by the analog watchdog	ADC_CR1 register control bits (x = don't care)		
	AWDSGL bit	AWDEN bit	JAWDEN bit
None	x	0	0
All injected channels	0	0	1
All regular channels	0	1	0
All regular and injected channels	0	1	1
Single ⁽¹⁾ injected channel	1	0	1
Single ⁽¹⁾ regular channel	1	1	0
Single ⁽¹⁾ regular or injected channel	1	1	1

1. Selected by the AWDCH[4:0] bits

13.3.8 Scan mode

This mode is used to scan a group of analog channels.

The Scan mode is selected by setting the SCAN bit in the ADC_CR1 register. Once this bit has been set, the ADC scans all the channels selected in the ADC_SQRx registers (for regular channels) or in the ADC_JSQR register (for injected channels). A single conversion is performed for each channel of the group. After each end of conversion, the next channel in the group is converted automatically. If the CONT bit is set, regular channel conversion does not stop at the last selected channel in the group but continues again from the first selected channel.

If the DMA bit is set, the direct memory access (DMA) controller is used to transfer the data converted from the regular group of channels (stored in the ADC_DR register) to SRAM after each regular channel conversion.

The EOC bit is set in the ADC_SR register:

- At the end of each regular group sequence if the EOCS bit is cleared to 0
- At the end of each regular channel conversion if the EOCS bit is set to 1

The data converted from an injected channel are always stored into the ADC_JDRx registers.

13.3.9 Injected channel management

Triggered injection

To use triggered injection, the JAUTO bit must be cleared in the ADC_CR1 register.

1. Start the conversion of a group of regular channels either by external trigger or by setting the SWSTART bit in the ADC_CR2 register.
2. If an external injected trigger occurs or if the JSWSTART bit is set during the conversion of a regular group of channels, the current conversion is reset and the injected channel sequence switches to Scan-once mode.
3. Then, the regular conversion of the regular group of channels is resumed from the last interrupted regular conversion.

If a regular event occurs during an injected conversion, the injected conversion is not interrupted but the regular sequence is executed at the end of the injected sequence.

Figure 47 shows the corresponding timing diagram.

Note: When using triggered injection, one must ensure that the interval between trigger events is longer than the injection sequence. For instance, if the sequence length is 30 ADC clock cycles (that is two conversions with a sampling time of 3 clock periods), the minimum interval between triggers must be 31 ADC clock cycles.

Auto-injection

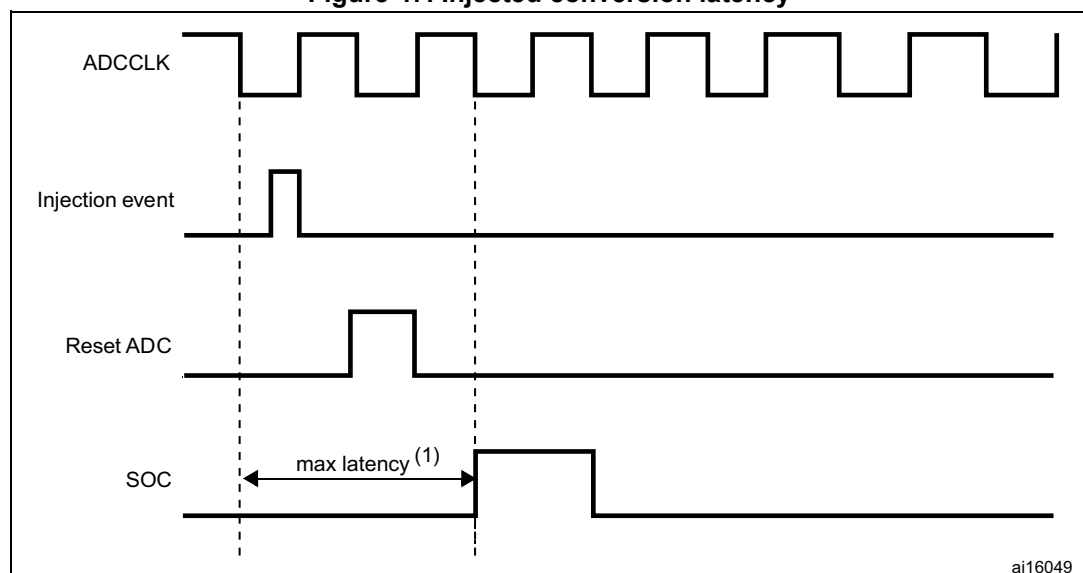
If the JAUTO bit is set, then the channels in the injected group are automatically converted after the regular group of channels. This can be used to convert a sequence of up to 20 conversions programmed in the ADC_SQRx and ADC_JSQR registers.

In this mode, external trigger on injected channels must be disabled.

If the CONT bit is also set in addition to the JAUTO bit, regular channels followed by injected channels are continuously converted.

Note: It is not possible to use both the auto-injected and discontinuous modes simultaneously.

Figure 47. Injected conversion latency



1. The maximum latency value can be found in the electrical characteristics of the STM32F40x and STM32F41x datasheets.

13.3.10 Discontinuous mode

Regular group

This mode is enabled by setting the DISCEN bit in the ADC_CR1 register. It can be used to convert a short sequence of n conversions ($n \leq 8$) that is part of the sequence of conversions selected in the ADC_SQRx registers. The value of n is specified by writing to the DISCNUM[2:0] bits in the ADC_CR1 register.

When an external trigger occurs, it starts the next n conversions selected in the ADC_SQRx registers until all the conversions in the sequence are done. The total sequence length is defined by the L[3:0] bits in the ADC_SQR1 register.

Example:

- $n = 3$, channels to be converted = 0, 1, 2, 3, 6, 7, 9, 10
- 1st trigger: sequence converted 0, 1, 2. An EOC event is generated at each conversion.
- 2nd trigger: sequence converted 3, 6, 7. An EOC event is generated at each conversion
- 3rd trigger: sequence converted 9, 10. An EOC event is generated at each conversion
- 4th trigger: sequence converted 0, 1, 2. An EOC event is generated at each conversion

Note: When a regular group is converted in discontinuous mode, no rollover occurs.

When all subgroups are converted, the next trigger starts the conversion of the first subgroup. In the example above, the 4th trigger reconverts the channels 0, 1 and 2 in the 1st subgroup.

Injected group

This mode is enabled by setting the JDISCEN bit in the ADC_CR1 register. It can be used to convert the sequence selected in the ADC_JSQR register, channel by channel, after an external trigger event.

When an external trigger occurs, it starts the next channel conversions selected in the ADC_JSQR registers until all the conversions in the sequence are done. The total sequence length is defined by the JL[1:0] bits in the ADC_JSQR register.

Example:

- $n = 1$, channels to be converted = 1, 2, 3
- 1st trigger: channel 1 converted
- 2nd trigger: channel 2 converted
- 3rd trigger: channel 3 converted and JEOP event generated
- 4th trigger: channel 1

Note: When all injected channels are converted, the next trigger starts the conversion of the first injected channel. In the example above, the 4th trigger reconverts the 1st injected channel 1.

It is not possible to use both the auto-injected and discontinuous modes simultaneously.

Discontinuous mode must not be set for regular and injected groups at the same time.

Discontinuous mode must be enabled only for the conversion of one group.

13.4 Data alignment

The ALIGN bit in the ADC_CR2 register selects the alignment of the data stored after conversion. Data can be right- or left-aligned as shown in [Figure 48](#) and [Figure 49](#).

The converted data value from the injected group of channels is decreased by the user-defined offset written in the ADC_JOFRx registers so the result can be a negative value. The SEXT bit represents the extended sign value.

For channels in a regular group, no offset is subtracted so only twelve bits are significant.

Figure 48. Right alignment of 12-bit data

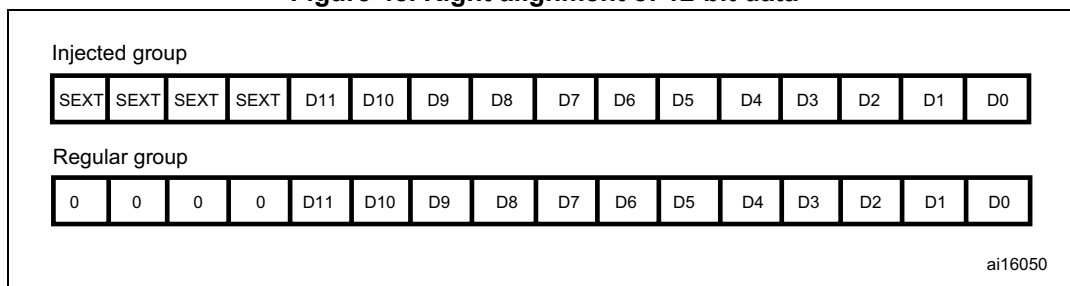
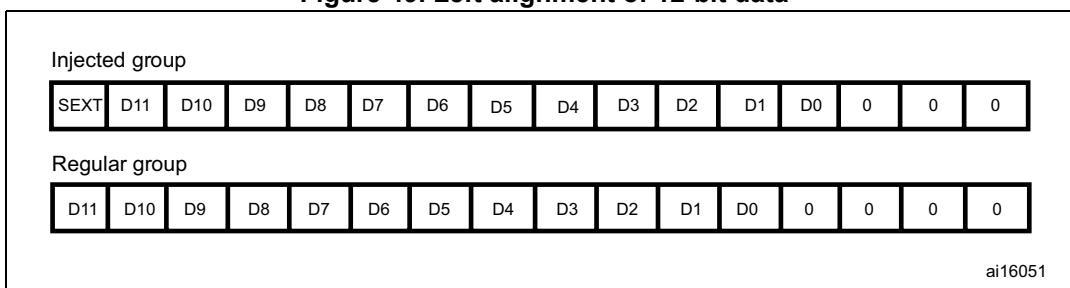
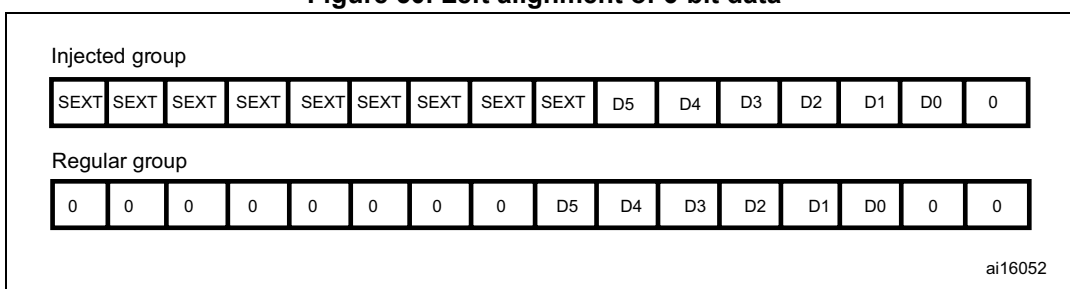


Figure 49. Left alignment of 12-bit data



Special case: when left-aligned, the data are aligned on a half-word basis except when the resolution is set to 6-bit. In that case, the data are aligned on a byte basis as shown in [Figure 50](#).

Figure 50. Left alignment of 6-bit data



13.5 Channel-wise programmable sampling time

The ADC samples the input voltage for a number of ADCCLK cycles that can be modified using the SMP[2:0] bits in the ADC_SMPR1 and ADC_SMPR2 registers. Each channel can be sampled with a different sampling time.

The total conversion time is calculated as follows:

$$T_{\text{conv}} = \text{Sampling time} + 12 \text{ cycles}$$

Example:

With ADCCLK = 30 MHz and sampling time = 3 cycles:

$$T_{\text{conv}} = 3 + 12 = 15 \text{ cycles} = 0.5 \mu\text{s with APB2 at 60 MHz}$$

13.6 Conversion on external trigger and trigger polarity

Conversion can be triggered by an external event (e.g. timer capture, EXTI line). If the EXTEN[1:0] control bits (for a regular conversion) or JEXTEN[1:0] bits (for an injected conversion) are different from “0b00”, then external events are able to trigger a conversion with the selected polarity. [Table 67](#) provides the correspondence between the EXTEN[1:0] and JEXTEN[1:0] values and the trigger polarity.

Table 67. Configuring the trigger polarity

Source	EXTEN[1:0] / JEXTEN[1:0]
Trigger detection disabled	00
Detection on the rising edge	01
Detection on the falling edge	10
Detection on both the rising and falling edges	11

Note: The polarity of the external trigger can be changed on the fly.

The EXTSEL[3:0] and JEXTSEL[3:0] control bits are used to select which out of 16 possible events can trigger conversion for the regular and injected groups.

[Table 68](#) gives the possible external trigger for regular conversion.

Table 68. External trigger for regular channels

Source	Type	EXTSEL[3:0]
TIM1_CH1 event	Internal signal from on-chip timers	0000
TIM1_CH2 event		0001
TIM1_CH3 event		0010
TIM2_CH2 event		0011
TIM2_CH3 event		0100
TIM2_CH4 event		0101
TIM2_TRGO event		0110
TIM3_CH1 event		0111
TIM3_TRGO event		1000
TIM4_CH4 event		1001
TIM5_CH1 event		1010
TIM5_CH2 event		1011
TIM5_CH3 event		1100
TIM8_CH1 event		1101
TIM8_TRGO event		1110
EXTI line11	External pin	1111

[Table 69](#) gives the possible external trigger for injected conversion.

Table 69. External trigger for injected channels

Source	Connection type	JEXTSEL[3:0]
TIM1_CH4 event	Internal signal from on-chip timers	0000
TIM1_TRGO event		0001
TIM2_CH1 event		0010
TIM2_TRGO event		0011
TIM3_CH2 event		0100
TIM3_CH4 event		0101
TIM4_CH1 event		0110
TIM4_CH2 event		0111
TIM4_CH3 event		1000
TIM4_TRGO event		1001
TIM5_CH4 event		1010
TIM5_TRGO event		1011
TIM8_CH2 event		1100
TIM8_CH3 event		1101
TIM8_CH4 event		1110
EXTI line15	External pin	1111

Software source trigger events can be generated by setting SWSTART (for regular conversion) or JSWSTART (for injected conversion) in ADC_CR2.

A regular group conversion can be interrupted by an injected trigger.

Note: The trigger selection can be changed on the fly. However, when the selection changes, there is a time frame of 1 APB clock cycle during which the trigger detection is disabled. This is to avoid spurious detection during transitions.

13.7 Fast conversion mode

It is possible to perform faster conversion by reducing the ADC resolution. The RES bits are used to select the number of bits available in the data register. The minimum conversion time for each resolution is then as follows:

- 12 bits: $3 + 12 = 15$ ADCCLK cycles
- 10 bits: $3 + 10 = 13$ ADCCLK cycles
- 8 bits: $3 + 8 = 11$ ADCCLK cycles
- 6 bits: $3 + 6 = 9$ ADCCLK cycles

13.8 Data management

13.8.1 Using the DMA

Since converted regular channel values are stored into a unique data register, it is useful to use DMA for conversion of more than one regular channel. This avoids the loss of the data already stored in the ADC_DR register.

When the DMA mode is enabled (DMA bit set to 1 in the ADC_CR2 register), after each conversion of a regular channel, a DMA request is generated. This allows the transfer of the converted data from the ADC_DR register to the destination location selected by the software.

Despite this, if data are lost (overflow), the OVR bit in the ADC_SR register is set and an interrupt is generated (if the OVRIE enable bit is set). DMA transfers are then disabled and DMA requests are no longer accepted. In this case, if a DMA request is made, the regular conversion in progress is aborted and further regular triggers are ignored. It is then necessary to clear the OVR flag and the DMAEN bit in the used DMA stream, and to re-initialize both the DMA and the ADC to have the wanted converted channel data transferred to the right memory location. Only then can the conversion be resumed and the data transfer, enabled again. Injected channel conversions are not impacted by overflow errors.

When OVR = 1 in DMA mode, the DMA requests are blocked after the last valid data have been transferred, which means that all the data transferred to the RAM can be considered as valid.

At the end of the last DMA transfer (number of transfers configured in the DMA controller's DMA_SxNDTR register):

- No new DMA request is issued to the DMA controller if the DDS bit is cleared to 0 in the ADC_CR2 register (this avoids generating an overflow error). However the DMA bit is not cleared by hardware. It must be written to 0, then to 1 to start a new transfer.
- Requests can continue to be generated if the DDS bit is set to 1. This allows configuring the DMA in double-buffer circular mode.

To recover the ADC from OVR state when the DMA is used, follow the steps below:

1. Reinitialize the DMA (adjust destination address and NDTR counter)
2. Clear the ADC OVR bit in ADC_SR register
3. Trigger the ADC to start the conversion.

13.8.2 Managing a sequence of conversions without using the DMA

If the conversions are slow enough, the conversion sequence can be handled by the software. In this case the EOCS bit must be set in the ADC_CR2 register for the EOC status bit to be set at the end of each conversion, and not only at the end of the sequence. When EOCS = 1, overflow detection is automatically enabled. Thus, each time a conversion is complete, EOC is set and the ADC_DR register can be read. The overflow management is the same as when the DMA is used.

To recover the ADC from OVR state when the EOCS is set, follow the steps below:

1. Clear the ADC OVR bit in ADC_SR register
2. Trigger the ADC to start the conversion.

13.8.3 Conversions without DMA and without overrun detection

It may be useful to let the ADC convert one or more channels without reading the data each time (if there is an analog watchdog for instance). For that, the DMA must be disabled (DMA = 0) and the EOC bit must be set at the end of a sequence only (EOCS = 0). In this configuration, overrun detection is disabled.

13.9 Multi ADC mode

In devices with two ADCs or more, the Dual (with two ADCs) and Triple (with three ADCs) ADC modes can be used (see [Figure 51](#)).

In multi ADC mode, the start of conversion is triggered alternately or simultaneously by the ADC1 master to the ADC2 and ADC3 slaves, depending on the mode selected by the MULTI[4:0] bits in the ADC_CCR register.

Note: In multi ADC mode, when configuring conversion trigger by an external event, the application must set trigger by the master only and disable trigger by slaves to prevent spurious triggers that would start unwanted slave conversions.

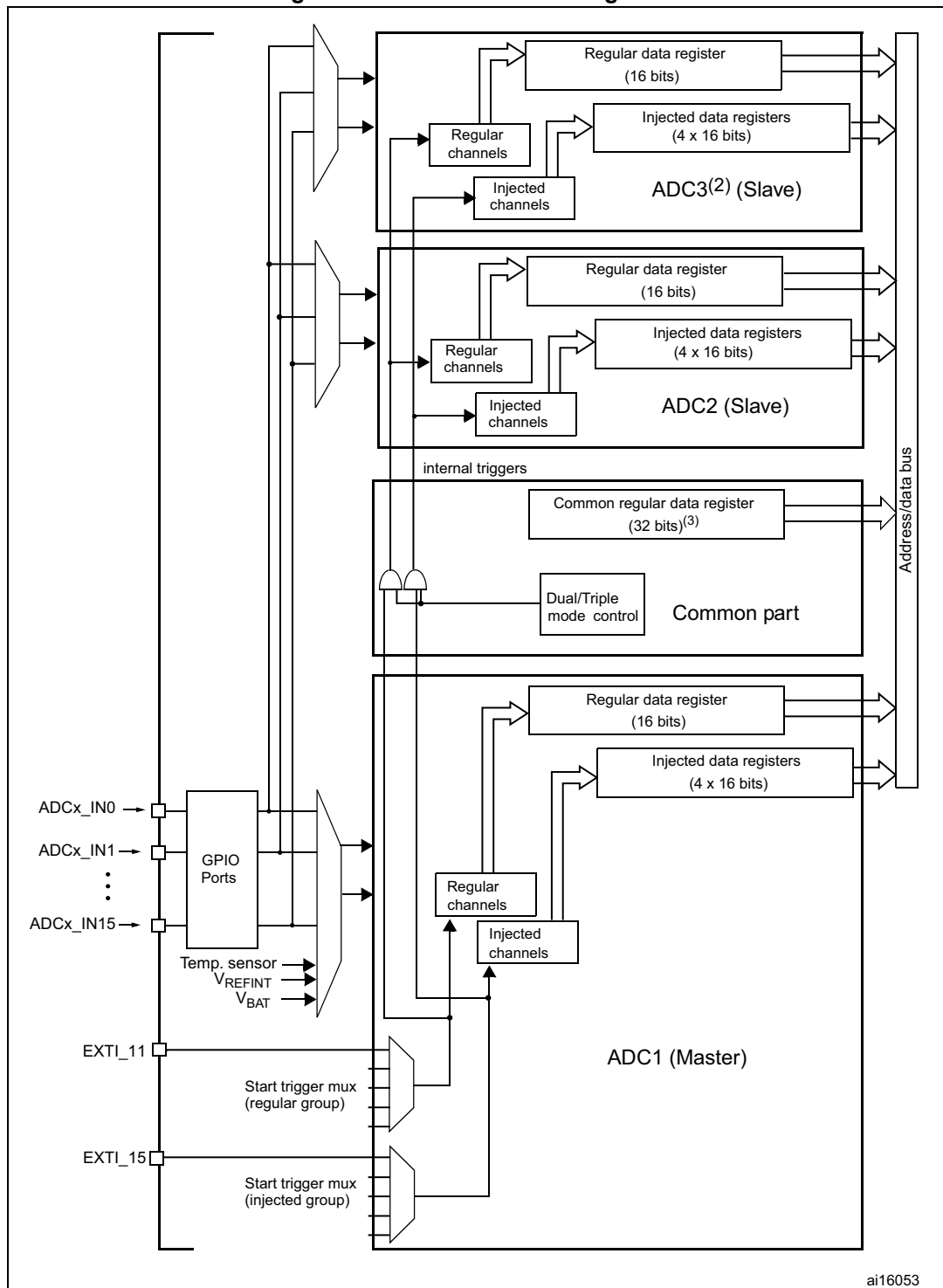
The four possible modes below are implemented:

- Injected simultaneous mode
- Regular simultaneous mode
- Interleaved mode
- Alternate trigger mode

It is also possible to use the previous modes combined in the following ways:

- Injected simultaneous mode + Regular simultaneous mode
- Regular simultaneous mode + Alternate trigger mode

Note: In multi ADC mode, the converted data can be read on the multi-mode data register (ADC_CDR). The status bits can be read in the multi-mode status register (ADC_CSR).

Figure 51. Multi ADC block diagram⁽¹⁾

1. Although external triggers are present on ADC2 and ADC3 they are not shown in this diagram.
2. In the Dual ADC mode, the ADC3 slave part is not present.
3. In Triple ADC mode, the ADC common data register (ADC_CDR) contains the ADC1, ADC2 and ADC3's regular converted data. All 32 register bits are used according to a selected storage order.
In Dual ADC mode, the ADC common data register (ADC_CDR) contains both the ADC1 and ADC2's regular converted data. All 32 register bits are used.

- DMA requests in Multi ADC mode:
In Multi ADC mode the DMA may be configured to transfer converted data in three different modes. In all cases, the DMA streams to use are those connected to the ADC:
 - **DMA mode 1:** On each DMA request (one data item is available), a half-word representing an ADC-converted data item is transferred.
In Triple ADC mode, ADC1 data are transferred on the first request, ADC2 data are transferred on the second request and ADC3 data are transferred on the third request; the sequence is repeated. So the DMA first transfers ADC1 data followed by ADC2 data followed by ADC3 data and so on.
DMA mode 1 is used in regular simultaneous triple mode only.
Example:
Regular simultaneous triple mode: 3 consecutive DMA requests are generated (one for each converted data item)
1st request: ADC_CDR[31:0] = ADC1_DR[15:0]
2nd request: ADC_CDR[31:0] = ADC2_DR[15:0]
3rd request: ADC_CDR[31:0] = ADC3_DR[15:0]
4th request: ADC_CDR[31:0] = ADC1_DR[15:0]
 - **DMA mode 2:** On each DMA request (two data items are available) two half-words representing two ADC-converted data items are transferred as a word.
In Dual ADC mode, both ADC2 and ADC1 data are transferred on the first request (ADC2 data take the upper half-word and ADC1 data take the lower half-word) and so on.
In Triple ADC mode, three DMA requests are generated. On the first request, both ADC2 and ADC1 data are transferred (ADC2 data take the upper half-word and ADC1 data take the lower half-word). On the second request, both ADC1 and ADC3 data are transferred (ADC1 data take the upper half-word and ADC3 data take the lower half-word). On the third request, both ADC3 and ADC2 data are transferred (ADC3 data take the upper half-word and ADC2 data take the lower half-word) and so on.
DMA mode 2 is used in interleaved mode and in regular simultaneous mode (for Dual ADC mode only).
Example:
 - a) Interleaved dual mode: a DMA request is generated each time 2 data items are available:
1st request: ADC_CDR[31:0] = ADC2_DR[15:0] | ADC1_DR[15:0]
2nd request: ADC_CDR[31:0] = ADC2_DR[15:0] | ADC1_DR[15:0]
 - b) Interleaved triple mode: a DMA request is generated each time 2 data items are available
1st request: ADC_CDR[31:0] = ADC2_DR[15:0] | ADC1_DR[15:0]
2nd request: ADC_CDR[31:0] = ADC1_DR[15:0] | ADC3_DR[15:0]
3rd request: ADC_CDR[31:0] = ADC3_DR[15:0] | ADC2_DR[15:0]
4th request: ADC_CDR[31:0] = ADC2_DR[15:0] | ADC1_DR[15:0]
 - **DMA mode 3:** This mode is similar to the DMA mode 2. The only differences are that the on each DMA request (two data items are available) two bytes

representing two ADC converted data items are transferred as a half-word. The data transfer order is similar to that of the DMA mode 2.

DMA mode 3 is used in interleaved mode in 6-bit and 8-bit resolutions (dual and triple mode).

Example:

- a) Interleaved dual mode: a DMA request is generated each time 2 data items are available
 1st request: `ADC_CDR[15:0] = ADC2_DR[7:0] | ADC1_DR[7:0]`
 2nd request: `ADC_CDR[15:0] = ADC2_DR[7:0] | ADC1_DR[7:0]`
- b) Interleaved triple mode: a DMA request is generated each time 2 data items are available
 1st request: `ADC_CDR[15:0] = ADC2_DR[7:0] | ADC1_DR[7:0]`
 2nd request: `ADC_CDR[15:0] = ADC1_DR[7:0] | ADC3_DR[7:0]`
 3rd request: `ADC_CDR[15:0] = ADC3_DR[7:0] | ADC2_DR[7:0]`
 4th request: `ADC_CDR[15:0] = ADC2_DR[7:0] | ADC1_DR[7:0]`

Overflow detection: If an overflow is detected on one of the concerned ADCs (ADC1 and ADC2 in dual and triple modes, ADC3 in triple mode only), the DMA requests are no longer issued to ensure that all the data transferred to the RAM are valid. It may happen that the EOC bit corresponding to one ADC remains set because the data register of this ADC contains valid data.

13.9.1 Injected simultaneous mode

This mode converts an injected group of channels. The external trigger source comes from the injected group multiplexer of ADC1 (selected by the JEXTSEL[3:0] bits in the ADC1_CR2 register). A simultaneous trigger is provided to ADC2 and ADC3.

Note: Do not convert the same channel on the two/three ADCs (no overlapping sampling times for the two/three ADCs when converting the same channel).

In simultaneous mode, one must convert sequences with the same length or ensure that the interval between triggers is longer than the longer of the 2 sequences (Dual ADC mode) /3 sequences (Triple ADC mode). Otherwise, the ADC with the shortest sequence may restart while the ADC with the longest sequence is completing the previous conversions.

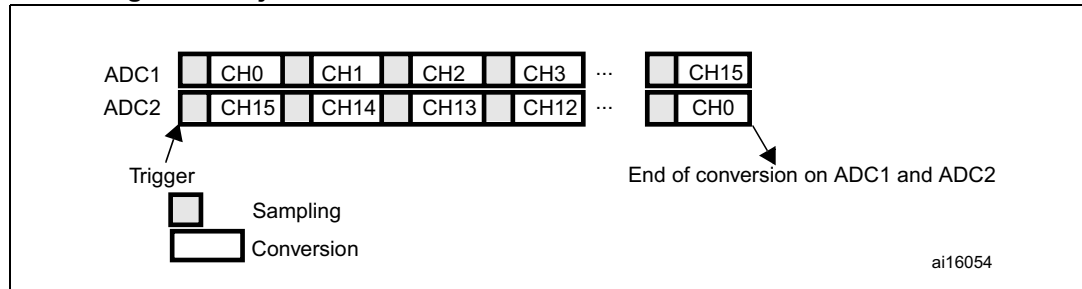
Regular conversions can be performed on one or all ADCs. In that case, they are independent of each other and are interrupted when an injected event occurs. They are resumed at the end of the injected conversion group.

Dual ADC mode

At the end of conversion event on ADC1 or ADC2:

- The converted data are stored into the ADC_JDRx registers of each ADC interface.
- A JEOC interrupt is generated (if enabled on one of the two ADC interfaces) when the ADC1/ADC2's injected channels have all been converted.

Figure 52. Injected simultaneous mode on 4 channels: dual ADC mode

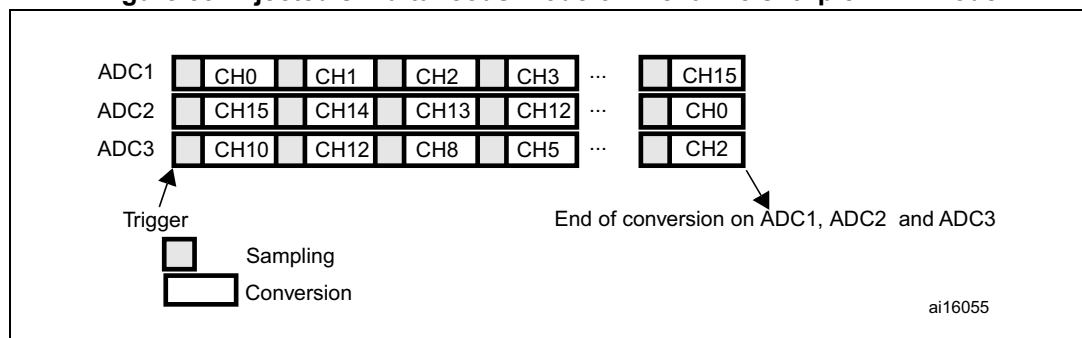


Triple ADC mode

At the end of conversion event on ADC1, ADC2 or ADC3:

- The converted data are stored into the ADC_JDRx registers of each ADC interface.
- A JEOC interrupt is generated (if enabled on one of the three ADC interfaces) when the ADC1/ADC2/ADC3's injected channels have all been converted.

Figure 53. Injected simultaneous mode on 4 channels: triple ADC mode



13.9.2 Regular simultaneous mode

This mode is performed on a regular group of channels. The external trigger source comes from the regular group multiplexer of ADC1 (selected by the EXTSEL[3:0] bits in the ADC1_CR2 register). A simultaneous trigger is provided to ADC2 and ADC3.

Note: *Do not convert the same channel on the two/three ADCs (no overlapping sampling times for the two/three ADCs when converting the same channel).*

In regular simultaneous mode, one must convert sequences with the same length or ensure that the interval between triggers is longer than the long conversion time of the 2 sequences (Dual ADC mode) / 3 sequences (Triple ADC mode). Otherwise, the ADC with the shortest sequence may restart while the ADC with the longest sequence is completing the previous conversions.

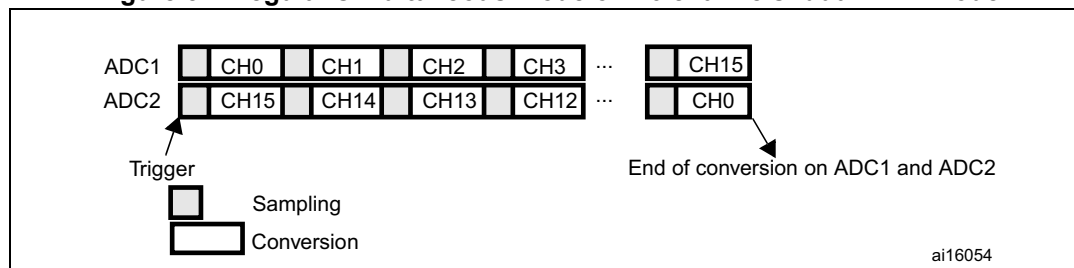
Injected conversions must be disabled.

Dual ADC mode

At the end of conversion event on ADC1 or ADC2:

- A 32-bit DMA transfer request is generated (if DMA[1:0] bits in the ADC_CCR register are equal to 0b10). This request transfers the ADC2 converted data stored in the upper half-word of the ADC_CDR 32-bit register to the SRAM and then the ADC1 converted data stored in the lower half-word of ADC_CDR to the SRAM.
- An EOC interrupt is generated (if enabled on one of the two ADC interfaces) when the ADC1/ADC2's regular channels have all been converted.

Figure 54. Regular simultaneous mode on 16 channels: dual ADC mode

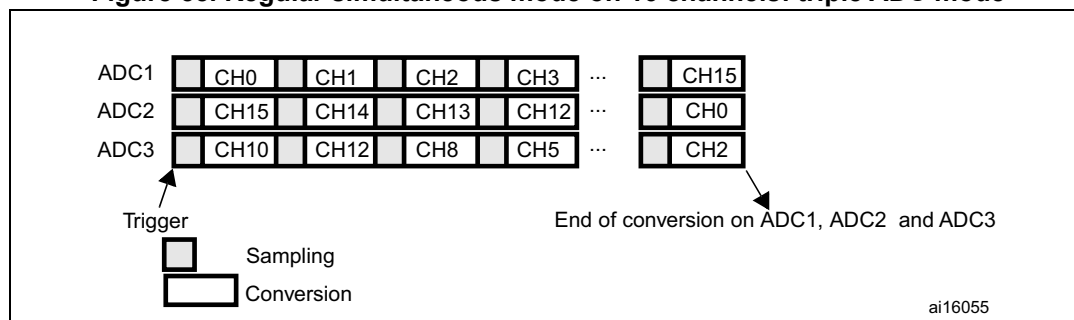


Triple ADC mode

At the end of conversion event on ADC1, ADC2 or ADC3:

- Three 32-bit DMA transfer requests are generated (if DMA[1:0] bits in the ADC_CCR register are equal to 0b01). Three transfers then take place from the ADC_CDR 32-bit register to SRAM: first the ADC1 converted data, then the ADC2 converted data and finally the ADC3 converted data. The process is repeated for each new three conversions.
- An EOC interrupt is generated (if enabled on one of the three ADC interfaces) when the ADC1/ADC2/ADC3's regular channels are have all been converted.

Figure 55. Regular simultaneous mode on 16 channels: triple ADC mode



13.9.3 Interleaved mode

This mode can be started only on a regular group (usually one channel). The external trigger source comes from the regular channel multiplexer of ADC1.

Dual ADC mode

After an external trigger occurs:

- ADC1 starts immediately
- ADC2 starts after a delay of several-ADC clock cycles

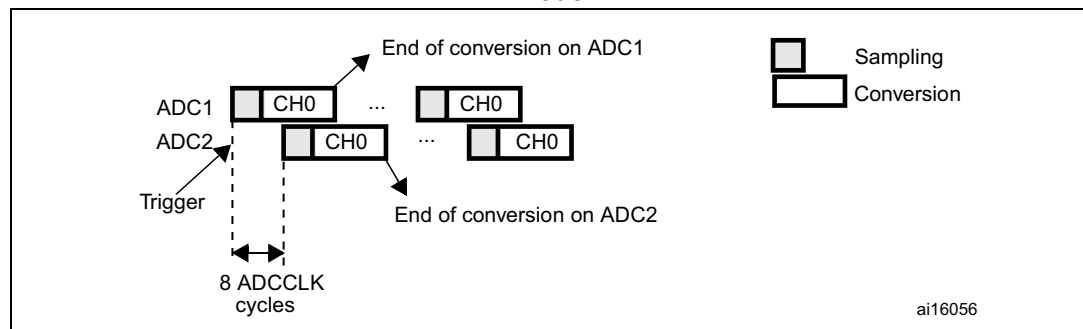
The minimum delay which separates 2 conversions in interleaved mode is configured in the DELAY bits in the ADC_CCR register. However, an ADC cannot start a conversion if the complementary ADC is still sampling its input (only one ADC can sample the input signal at a given time). In this case, the delay becomes the sampling time + 2 ADC clock cycles. For instance, if DELAY = 5 clock cycles and the sampling takes 15 clock cycles on both ADCs, then 17 clock cycles separate conversions on ADC1 and ADC2).

If the CONT bit is set on both ADC1 and ADC2, the selected regular channels of both ADCs are continuously converted.

Note: *If the conversion sequence is interrupted (for instance when DMA end of transfer occurs), the multi-ADC sequencer must be reset by configuring it in independent mode first (bits DUAL[4:0] = 00000) before reprogramming the interleaved mode.*

After an EOC interrupt is generated by ADC2 (if enabled through the EOCIE bit) a 32-bit DMA transfer request is generated (if the DMA[1:0] bits in ADC_CCR are equal to 0b10). This request first transfers the ADC2 converted data stored in the upper half-word of the ADC_CDR 32-bit register into SRAM, then the ADC1 converted data stored in the register's lower half-word into SRAM.

Figure 56. Interleaved mode on 1 channel in continuous conversion mode: dual ADC mode



Triple ADC mode

After an external trigger occurs:

- ADC1 starts immediately and
- ADC2 starts after a delay of several ADC clock cycles
- ADC3 starts after a delay of several ADC clock cycles referred to the ADC2 conversion

The minimum delay which separates 2 conversions in interleaved mode is configured in the DELAY bits in the ADC_CCR register. However, an ADC cannot start a conversion if the complementary ADC is still sampling its input (only one ADC can sample the input signal at

a given time). In this case, the delay becomes the sampling time + 2 ADC clock cycles. For instance, if DELAY = 5 clock cycles and the sampling takes 15 clock cycles on the three ADCs, then 17 clock cycles separate the conversions on ADC1, ADC2 and ADC3).

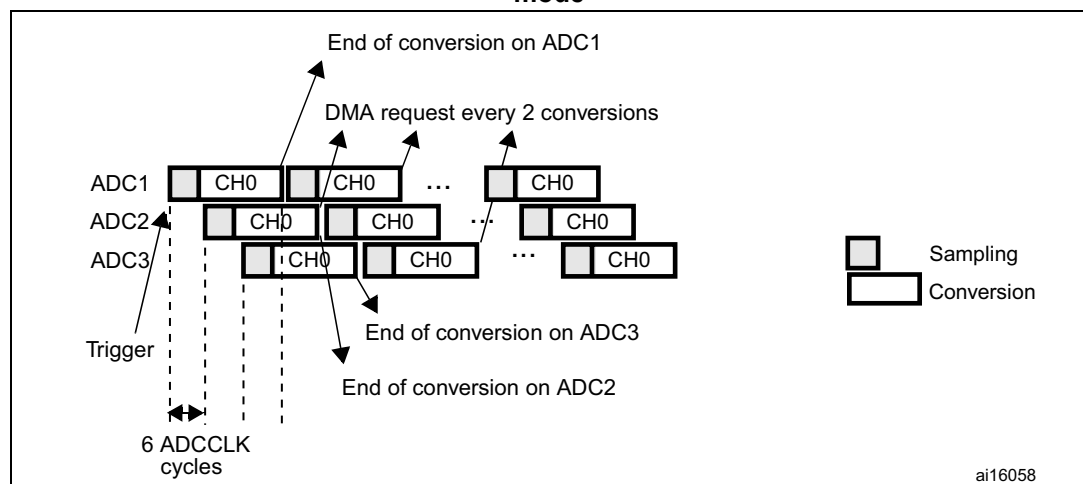
If the CONT bit is set on ADC1, ADC2 and ADC3, the selected regular channels of all ADCs are continuously converted.

Note: *If the conversion sequence is interrupted (for instance when DMA end of transfer occurs), the multi-ADC sequencer must be reset by configuring it in independent mode first (bits DUAL[4:0] = 00000) before reprogramming the interleaved mode.*

In this mode a DMA request is generated each time 2 data items are available, (if the DMA[1:0] bits in the ADC_CCR register are equal to 0b10). The request first transfers the first converted data stored in the lower half-word of the ADC_CDR 32-bit register to SRAM, then it transfers the second converted data stored in ADC_CDR's upper half-word to SRAM. The sequence is the following:

- 1st request: ADC_CDR[31:0] = ADC2_DR[15:0] | ADC1_DR[15:0]
- 2nd request: ADC_CDR[31:0] = ADC1_DR[15:0] | ADC3_DR[15:0]
- 3rd request: ADC_CDR[31:0] = ADC3_DR[15:0] | ADC2_DR[15:0]
- 4th request: ADC_CDR[31:0] = ADC2_DR[15:0] | ADC1_DR[15:0], ...

Figure 57. Interleaved mode on 1 channel in continuous conversion mode: triple ADC mode



13.9.4 Alternate trigger mode

This mode can be started only on an injected group. The source of external trigger comes from the injected group multiplexer of ADC1.

Note: *Regular conversions can be enabled on one or all ADCs. In this case the regular conversions are independent of each other. A regular conversion is interrupted when the*

ADC has to perform an injected conversion. It is resumed when the injected conversion is finished.

If the conversion sequence is interrupted (for instance when DMA end of transfer occurs), the multi-ADC sequencer must be reset by configuring it in independent mode first (bits $DUAL[4:0] = 00000$) before reprogramming the interleaved mode.

The time interval between 2 trigger events must be greater than or equal to 1 ADC clock period. The minimum time interval between 2 trigger events that start conversions on the same ADC is the same as in the single ADC mode.

Dual ADC mode

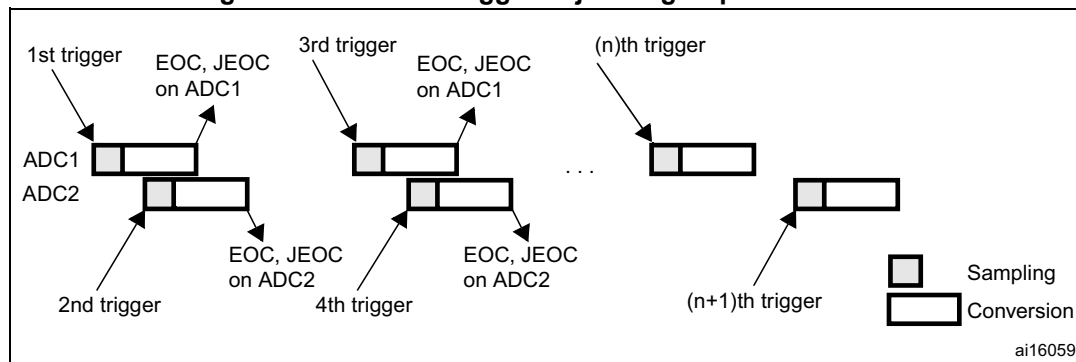
- When the 1st trigger occurs, all injected ADC1 channels in the group are converted
- When the 2nd trigger occurs, all injected ADC2 channels in the group are converted
- and so on

A JEOC interrupt, if enabled, is generated after all injected ADC1 channels in the group have been converted.

A JEOC interrupt, if enabled, is generated after all injected ADC2 channels in the group have been converted.

If another external trigger occurs after all injected channels in the group have been converted then the alternate trigger process restarts by converting the injected ADC1 channels in the group.

Figure 58. Alternate trigger: injected group of each ADC



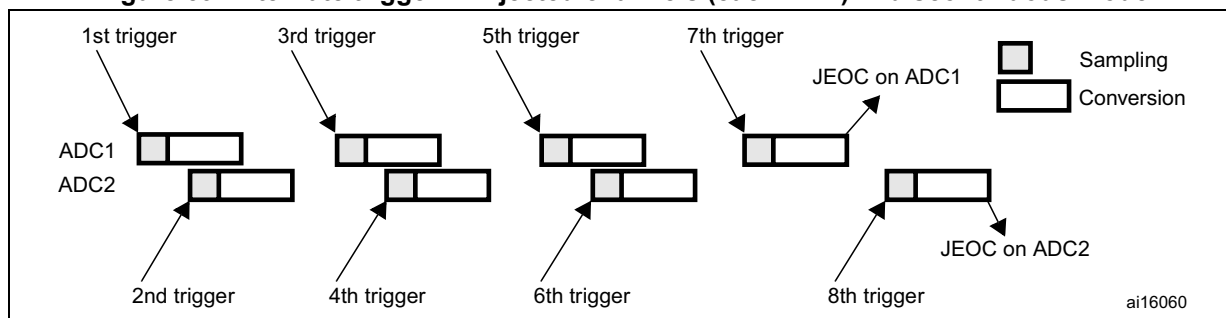
If the injected discontinuous mode is enabled for both ADC1 and ADC2:

- When the 1st trigger occurs, the first injected ADC1 channel is converted.
- When the 2nd trigger occurs, the first injected ADC2 channel are converted
- and so on

A JEOC interrupt, if enabled, is generated after all injected ADC1 channels in the group have been converted.

A JEOC interrupt, if enabled, is generated after all injected ADC2 channels in the group have been converted.

If another external trigger occurs after all injected channels in the group have been converted then the alternate trigger process restarts.

Figure 59. Alternate trigger: 4 injected channels (each ADC) in discontinuous mode**Triple ADC mode**

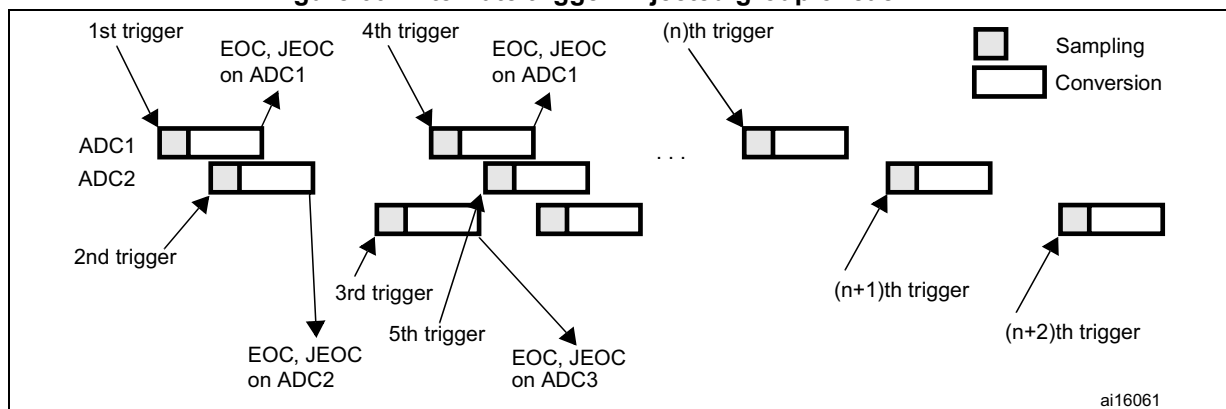
- When the 1st trigger occurs, all injected ADC1 channels in the group are converted.
- When the 2nd trigger occurs, all injected ADC2 channels in the group are converted.
- When the 3rd trigger occurs, all injected ADC3 channels in the group are converted.
- and so on

A JEOC interrupt, if enabled, is generated after all injected ADC1 channels in the group have been converted.

A JEOC interrupt, if enabled, is generated after all injected ADC2 channels in the group have been converted.

A JEOC interrupt, if enabled, is generated after all injected ADC3 channels in the group have been converted.

If another external trigger occurs after all injected channels in the group have been converted then the alternate trigger process restarts by converting the injected ADC1 channels in the group.

Figure 60. Alternate trigger: injected group of each ADC**13.9.5 Combined regular/injected simultaneous mode**

It is possible to interrupt the simultaneous conversion of a regular group to start the simultaneous conversion of an injected group.

Note: *In combined regular/injected simultaneous mode, one must convert sequences with the same length or ensure that the interval between triggers is longer than the long conversion time of the 2 sequences (Dual ADC mode) / 3 sequences (Triple ADC mode). Otherwise, the*

ADC with the shortest sequence may restart while the ADC with the longest sequence is completing the previous conversions.

13.9.6 Combined regular simultaneous + alternate trigger mode

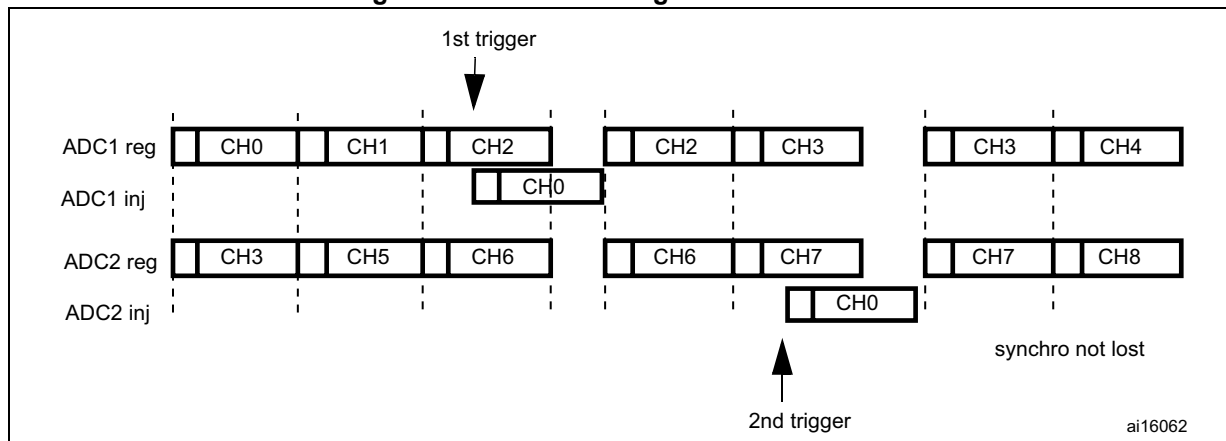
It is possible to interrupt the simultaneous conversion of a regular group to start the alternate trigger conversion of an injected group. [Figure 61](#) shows the behavior of an alternate trigger interrupting a simultaneous regular conversion.

The injected alternate conversion is immediately started after the injected event. If regular conversion is already running, in order to ensure synchronization after the injected conversion, the regular conversion of all (master/slave) ADCs is stopped and resumed synchronously at the end of the injected conversion.

Note: *In combined regular simultaneous + alternate trigger mode, one must convert sequences with the same length or ensure that the interval between triggers is longer than the long conversion time of the 2 sequences (Dual ADC mode) / 3 sequences (Triple ADC mode). Otherwise, the ADC with the shortest sequence may restart while the ADC with the longest sequence is completing the previous conversions.*

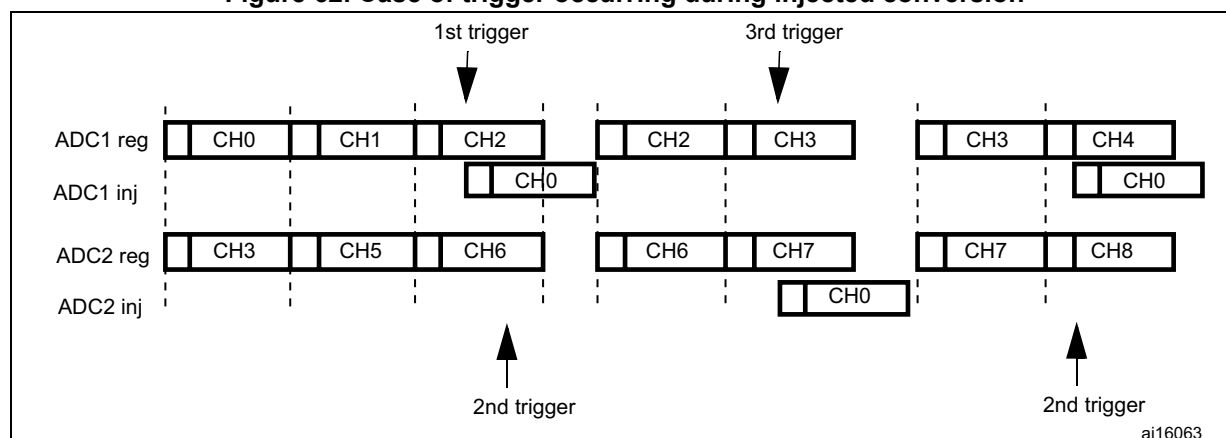
If the conversion sequence is interrupted (for instance when DMA end of transfer occurs), the multi-ADC sequencer must be reset by configuring it in independent mode first (bits DUAL[4:0] = 00000) before reprogramming the interleaved mode.

Figure 61. Alternate + regular simultaneous



If a trigger occurs during an injected conversion that has interrupted a regular conversion, it is ignored. [Figure 62](#) shows the behavior in this case (2nd trigger is ignored).

Figure 62. Case of trigger occurring during injected conversion



13.10 Temperature sensor

The temperature sensor can be used to measure the junction temperature (T_J) of the device.

- On STM32F40x and STM32F41x devices, the temperature sensor is internally connected to ADC1_IN16 channel which is used to convert the sensor output voltage to a digital value.
- On STM32F42x and STM32F43x devices, the temperature sensor is internally connected to the same input channel, ADC1_IN18, as VBAT: ADC1_IN18 is used to convert the sensor output voltage or VBAT into a digital value. Only one conversion, temperature sensor or VBAT, must be selected at a time. When the temperature sensor and the VBAT conversion are set simultaneously, only the VBAT conversion is performed.

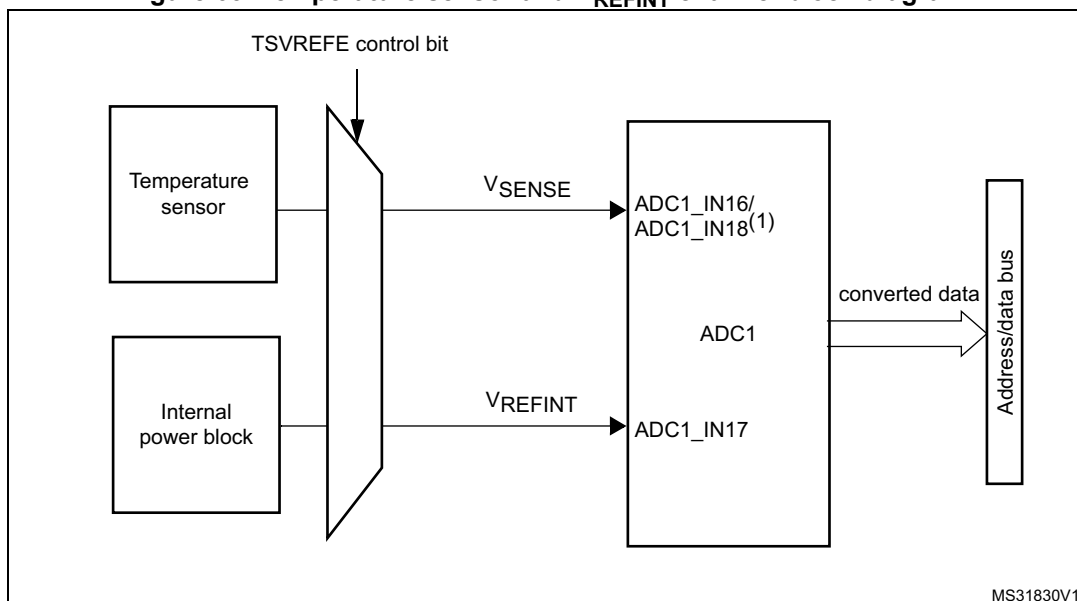
Figure 63 shows the block diagram of the temperature sensor.

When not in use, the sensor can be put in power down mode.

Note: The *TSVREFE* bit must be set to enable the conversion of both internal channels: the ADC1_IN16 or ADC1_IN18 (temperature sensor) and the ADC1_IN17 (VREFINT).

Main features

- Supported temperature range: -40 to 125 °C
- Precision: ± 1.5 °C

Figure 63. Temperature sensor and V_{REFINT} channel block diagram

1. V_{SENSE} is input to ADC1_IN16 for the STM23F40x and STM32F41x devices and to ADC1_IN18 for the STM32F42x and STM32F43x devices.

Reading the temperature

To use the sensor:

3. Select ADC1_IN16 or ADC1_IN18 input channel.
4. Select a sampling time greater than the minimum sampling time specified in the datasheet.
5. Set the TSVREFE bit in the ADC_CCR register to wake up the temperature sensor from power down mode
6. Start the ADC conversion by setting the SWSTART bit (or by external trigger)
7. Read the resulting V_{SENSE} data in the ADC data register
8. Calculate the temperature using the following formula:

$$\text{Temperature (in } ^\circ\text{C)} = \{(V_{\text{SENSE}} - V_{25}) / \text{Avg_Slope}\} + 25$$

Where:

 - V₂₅ = V_{SENSE} value for 25° C
 - Avg_Slope = average slope of the temperature vs. V_{SENSE} curve (given in mV/°C or μV/°C)

Refer to the datasheet's electrical characteristics section for the actual values of V₂₅ and Avg_Slope.

Note: *The sensor has a startup time after waking from power down mode before it can output V_{SENSE} at the correct level. The ADC also has a startup time after power-on, so to minimize the delay, the ADON and TSVREFE bits should be set at the same time.*

The temperature sensor output voltage changes linearly with temperature. The offset of this linear function depends on each chip due to process variation (up to 45 °C from one chip to another).

The internal temperature sensor is more suited for applications that detect temperature variations instead of absolute temperatures. If accurate temperature reading is required, an external temperature sensor should be used.

13.11 Battery charge monitoring

The VBATE bit in the ADC_CCR register is used to switch to the battery voltage. As the V_{BAT} voltage could be higher than V_{DDA} , to ensure the correct operation of the ADC, the V_{BAT} pin is internally connected to a bridge divider.

When the VBATE is set, the bridge is automatically enabled to connect:

- $V_{BAT}/2$ to the ADC1_IN18 input channel, on STM32F40xx and STM32F41xx devices
- $V_{BAT}/4$ to the ADC1_IN18 input channel, on STM32F42xx and STM32F43xx devices

Note: On STM32F42xx and STM32F43xx devices, VBAT and temperature sensor are connected to the same ADC internal channel (ADC1_IN18). Only one conversion, either temperature sensor or VBAT, must be selected at a time. When both conversion are enabled simultaneously, only the VBAT conversion is performed.

13.12 ADC interrupts

An interrupt can be produced on the end of conversion for regular and injected groups, when the analog watchdog status bit is set and when the overrun status bit is set. Separate interrupt enable bits are available for flexibility.

Two other flags are present in the ADC_SR register, but there is no interrupt associated with them:

- JSTRT (Start of conversion for channels of an injected group)
- STRT (Start of conversion for channels of a regular group)

Table 70. ADC interrupts

Interrupt event	Event flag	Enable control bit
End of conversion of a regular group	EOC	EOCIE
End of conversion of an injected group	JEOC	JEOCIE
Analog watchdog status bit is set	AWD	AWDIE
Overrun	OVR	OVRIE

13.13 ADC registers

Refer to [Section 1.1: List of abbreviations for registers](#) for registers for a list of abbreviations used in register descriptions.

The peripheral registers must be written at word level (32 bits). Read accesses can be done by bytes (8 bits), half-words (16 bits) or words (32 bits).

13.13.1 ADC status register (ADC_SR)

Address offset: 0x00

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										OVR	STRT	JSTRT	JEOC	EOC	AWD
										rc_w0	rc_w0	rc_w0	rc_w0	rc_w0	rc_w0

Bits 31:6 Reserved, must be kept at reset value.

Bit 5 OVR: Overrun

This bit is set by hardware when data are lost (either in single mode or in dual/triple mode). It is cleared by software. Overrun detection is enabled only when DMA = 1 or EOCS = 1.

0: No overrun occurred
1: Overrun has occurred

Bit 4 STRT: Regular channel start flag

This bit is set by hardware when regular channel conversion starts. It is cleared by software.

0: No regular channel conversion started
1: Regular channel conversion has started

Bit 3 JSTRT: Injected channel start flag

This bit is set by hardware when injected group conversion starts. It is cleared by software.

0: No injected group conversion started
1: Injected group conversion has started

Bit 2 JEOC: Injected channel end of conversion

This bit is set by hardware at the end of the conversion of all injected channels in the group. It is cleared by software.

0: Conversion is not complete
1: Conversion complete

Bit 1 EOC: Regular channel end of conversion

This bit is set by hardware at the end of the conversion of a regular group of channels. It is cleared by software or by reading the ADC_DR register.

0: Conversion not complete (EOCS=1), or sequence of conversions not complete (EOCS=0)
1: Conversion complete (EOCS=1), or sequence of conversions complete (EOCS=0)

Bit 0 AWD: Analog watchdog flag

This bit is set by hardware when the converted voltage crosses the values programmed in the ADC_LTR and ADC_HTR registers. It is cleared by software.

0: No analog watchdog event occurred
1: Analog watchdog event occurred

13.13.2 ADC control register 1 (ADC_CR1)

Address offset: 0x04

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					OVRIE	RES		AWDEN	JAWDEN	Reserved					
					rw	rw	rw	rw	rw						
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DISCNUM[2:0]			JDISCEN	DISCEN	JAUTO	AWDSGL	SCAN	JEOCIE	AWDIE	EOCIE	AWDCH[4:0]				
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:27 Reserved, must be kept at reset value.

Bit 26 **OVRIE**: Overrun interrupt enable

This bit is set and cleared by software to enable/disable the Overrun interrupt.

0: Overrun interrupt disabled

1: Overrun interrupt enabled. An interrupt is generated when the OVR bit is set.

Bits 25:24 **RES[1:0]**: Resolution

These bits are written by software to select the resolution of the conversion.

00: 12-bit (15 ADCCLK cycles)

01: 10-bit (13 ADCCLK cycles)

10: 8-bit (11 ADCCLK cycles)

11: 6-bit (9 ADCCLK cycles)

Bit 23 **AWDEN**: Analog watchdog enable on regular channels

This bit is set and cleared by software.

0: Analog watchdog disabled on regular channels

1: Analog watchdog enabled on regular channels

Bit 22 **JAWDEN**: Analog watchdog enable on injected channels

This bit is set and cleared by software.

0: Analog watchdog disabled on injected channels

1: Analog watchdog enabled on injected channels

Bits 21:16 Reserved, must be kept at reset value.

Bits 15:13 **DISCNUM[2:0]**: Discontinuous mode channel count

These bits are written by software to define the number of regular channels to be converted in discontinuous mode, after receiving an external trigger.

000: 1 channel

001: 2 channels

...

111: 8 channels

Bit 12 **JDISCEN**: Discontinuous mode on injected channels

This bit is set and cleared by software to enable/disable discontinuous mode on the injected channels of a group.

0: Discontinuous mode on injected channels disabled

1: Discontinuous mode on injected channels enabled

- Bit 11 **DISCEN**: Discontinuous mode on regular channels
 This bit is set and cleared by software to enable/disable Discontinuous mode on regular channels.
 0: Discontinuous mode on regular channels disabled
 1: Discontinuous mode on regular channels enabled
- Bit 10 **JAUTO**: Automatic injected group conversion
 This bit is set and cleared by software to enable/disable automatic injected group conversion after regular group conversion.
 0: Automatic injected group conversion disabled
 1: Automatic injected group conversion enabled
- Bit 9 **AWDSGL**: Enable the watchdog on a single channel in scan mode
 This bit is set and cleared by software to enable/disable the analog watchdog on the channel identified by the AWDCH[4:0] bits.
 0: Analog watchdog enabled on all channels
 1: Analog watchdog enabled on a single channel
- Bit 8 **SCAN**: Scan mode
 This bit is set and cleared by software to enable/disable the Scan mode. In Scan mode, the inputs selected through the ADC_SQRx or ADC_JSQRx registers are converted.
 0: Scan mode disabled
 1: Scan mode enabled
Note: An EOC interrupt is generated if the EOCIE bit is set:
 – At the end of each regular group sequence if the EOCS bit is cleared to 0
 – At the end of each regular channel conversion if the EOCS bit is set to 1
Note: A JEOP interrupt is generated only on the end of conversion of the last channel if the JEOPIE bit is set.
- Bit 7 **JEOPIE**: Interrupt enable for injected channels
 This bit is set and cleared by software to enable/disable the end of conversion interrupt for injected channels.
 0: JEOP interrupt disabled
 1: JEOP interrupt enabled. An interrupt is generated when the JEOP bit is set.
- Bit 6 **AWDIE**: Analog watchdog interrupt enable
 This bit is set and cleared by software to enable/disable the analog watchdog interrupt.
 0: Analog watchdog interrupt disabled
 1: Analog watchdog interrupt enabled
- Bit 5 **EOCIE**: Interrupt enable for EOC
 This bit is set and cleared by software to enable/disable the end of conversion interrupt.
 0: EOC interrupt disabled
 1: EOC interrupt enabled. An interrupt is generated when the EOC bit is set.
- Bits 4:0 **AWDCH[4:0]**: Analog watchdog channel select bits
 These bits are set and cleared by software. They select the input channel to be guarded by the analog watchdog.
Note: 00000: ADC analog input Channel0
 00001: ADC analog input Channel1
 ...
 01111: ADC analog input Channel15
 10000: ADC analog input Channel16
 10001: ADC analog input Channel17
 10010: ADC analog input Channel18
 Other values reserved

13.13.3 ADC control register 2 (ADC_CR2)

Address offset: 0x08

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
reserved	SWST ART	EXTEN		EXTSEL[3:0]				reserved	JSWST ART	JEXTEN		JEXTSEL[3:0]			
	rw	rw	rw	rw	rw	rw	rw		rw	rw	rw	rw	rw	rw	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
reserved				ALIGN	EOCS	DDS	DMA	Reserved						CONT	ADON
				rw	rw	rw	rw							rw	rw

Bit 31 Reserved, must be kept at reset value.

Bit 30 **SWSTART**: Start conversion of regular channels

This bit is set by software to start conversion and cleared by hardware as soon as the conversion starts.

0: Reset state

1: Starts conversion of regular channels

Note: This bit can be set only when ADON = 1 otherwise no conversion is launched.

Bits 29:28 **EXTEN**: External trigger enable for regular channels

These bits are set and cleared by software to select the external trigger polarity and enable the trigger of a regular group.

00: Trigger detection disabled

01: Trigger detection on the rising edge

10: Trigger detection on the falling edge

11: Trigger detection on both the rising and falling edges

Bits 27:24 **EXTSEL[3:0]**: External event select for regular group

These bits select the external event used to trigger the start of conversion of a regular group:

0000: Timer 1 CC1 event

0001: Timer 1 CC2 event

0010: Timer 1 CC3 event

0011: Timer 2 CC2 event

0100: Timer 2 CC3 event

0101: Timer 2 CC4 event

0110: Timer 2 TRGO event

0111: Timer 3 CC1 event

1000: Timer 3 TRGO event

1001: Timer 4 CC4 event

1010: Timer 5 CC1 event

1011: Timer 5 CC2 event

1100: Timer 5 CC3 event

1101: Timer 8 CC1 event

1110: Timer 8 TRGO event

1111: EXTI line11

Bit 23 Reserved, must be kept at reset value.

Bit 22 **JSWSTART**: Start conversion of injected channels

This bit is set by software and cleared by hardware as soon as the conversion starts.

0: Reset state

1: Starts conversion of injected channels

Note: This bit can be set only when ADON = 1 otherwise no conversion is launched.

Bits 21:20 **JEXTEN**: External trigger enable for injected channels

These bits are set and cleared by software to select the external trigger polarity and enable the trigger of an injected group.

00: Trigger detection disabled

01: Trigger detection on the rising edge

10: Trigger detection on the falling edge

11: Trigger detection on both the rising and falling edges

Bits 19:16 **JEXTSEL[3:0]**: External event select for injected group

These bits select the external event used to trigger the start of conversion of an injected group.

0000: Timer 1 CC4 event

0001: Timer 1 TRGO event

0010: Timer 2 CC1 event

0011: Timer 2 TRGO event

0100: Timer 3 CC2 event

0101: Timer 3 CC4 event

0110: Timer 4 CC1 event

0111: Timer 4 CC2 event

1000: Timer 4 CC3 event

1001: Timer 4 TRGO event

1010: Timer 5 CC4 event

1011: Timer 5 TRGO event

1100: Timer 8 CC2 event

1101: Timer 8 CC3 event

1110: Timer 8 CC4 event

1111: EXTI line15

Bits 15:12 Reserved, must be kept at reset value.

Bit 11 **ALIGN**: Data alignment

This bit is set and cleared by software. Refer to [Figure 48](#) and [Figure 49](#).

0: Right alignment

1: Left alignment

Bit 10 **EOCS**: End of conversion selection

This bit is set and cleared by software.

0: The EOC bit is set at the end of each sequence of regular conversions. Overrun detection is enabled only if DMA=1.

1: The EOC bit is set at the end of each regular conversion. Overrun detection is enabled.

Bit 9 **DDS**: DMA disable selection (for single ADC mode)

This bit is set and cleared by software.

0: No new DMA request is issued after the last transfer (as configured in the DMA controller)

1: DMA requests are issued as long as data are converted and DMA=1

Bit 8 **DMA**: Direct memory access mode (for single ADC mode)

This bit is set and cleared by software. Refer to the DMA controller chapter for more details.

0: DMA mode disabled

1: DMA mode enabled

Bits 7:2 Reserved, must be kept at reset value.

Bit 1 **CONT**: Continuous conversion

This bit is set and cleared by software. If it is set, conversion takes place continuously until it is cleared.

0: Single conversion mode

1: Continuous conversion mode

Bit 0 **ADON**: A/D Converter ON / OFF

This bit is set and cleared by software.

Note: 0: Disable ADC conversion and go to power down mode

1: Enable ADC

13.13.4 ADC sample time register 1 (ADC_SMPR1)

Address offset: 0x0C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					SMP18[2:0]			SMP17[2:0]			SMP16[2:0]			SMP15[2:1]	
					rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SMP15_0	SMP14[2:0]			SMP13[2:0]			SMP12[2:0]			SMP11[2:0]			SMP10[2:0]		
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31: 27 Reserved, must be kept at reset value.

Bits 26:0 **SMPx[2:0]**: Channel x sampling time selection

These bits are written by software to select the sampling time individually for each channel.

During sampling cycles, the channel selection bits must remain unchanged.

Note: 000: 3 cycles

001: 15 cycles

010: 28 cycles

011: 56 cycles

100: 84 cycles

101: 112 cycles

110: 144 cycles

111: 480 cycles

13.13.5 ADC sample time register 2 (ADC_SMPR2)

Address offset: 0x10

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		SMP9[2:0]			SMP8[2:0]			SMP7[2:0]			SMP6[2:0]			SMP5[2:1]	
		rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SMP5_0	SMP4[2:0]			SMP3[2:0]			SMP2[2:0]			SMP1[2:0]			SMP0[2:0]		
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:30 Reserved, must be kept at reset value.

Bits 29:0 **SMPx[2:0]**: Channel x sampling time selection

These bits are written by software to select the sampling time individually for each channel.
During sample cycles, the channel selection bits must remain unchanged.

Note: 000: 3 cycles
001: 15 cycles
010: 28 cycles
011: 56 cycles
100: 84 cycles
101: 112 cycles
110: 144 cycles
111: 480 cycles

13.13.6 ADC injected channel data offset register x (ADC_JOFRx) (x=1..4)

Address offset: 0x14-0x20

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				JOFFSETx[11:0]											
				rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:12 Reserved, must be kept at reset value.

Bits 11:0 **JOFFSETx[11:0]**: Data offset for injected channel x

These bits are written by software to define the offset to be subtracted from the raw converted data when converting injected channels. The conversion result can be read from in the ADC_JDRx registers.

13.13.7 ADC watchdog higher threshold register (ADC_HTR)

Address offset: 0x24

Reset value: 0x0000 0FFF

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				HT[11:0]											
				rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:12 Reserved, must be kept at reset value.

Bits 11:0 **HT[11:0]**: Analog watchdog higher threshold

These bits are written by software to define the higher threshold for the analog watchdog.

Note: *The software can write to these registers when an ADC conversion is ongoing. The programmed value is effective when the next conversion is complete. Writing to this register is performed with a write delay that can create uncertainty on the effective time at which the new value is programmed.*

13.13.8 ADC watchdog lower threshold register (ADC_LTR)

Address offset: 0x28

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				LT[11:0]											
				rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:12 Reserved, must be kept at reset value.

Bits 11:0 **LT[11:0]**: Analog watchdog lower threshold

These bits are written by software to define the lower threshold for the analog watchdog.

Note: *The software can write to these registers when an ADC conversion is ongoing. The programmed value is effective when the next conversion is complete. Writing to this register is performed with a write delay that can create uncertainty on the effective time at which the new value is programmed.*

13.13.9 ADC regular sequence register 1 (ADC_SQR1)

Address offset: 0x2C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								L[3:0]				SQ16[4:1]			
								rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SQ16_0	SQ15[4:0]					SQ14[4:0]					SQ13[4:0]				
rw	rw	rw	rw	rw	rw	rw	rw				rw	rw	rw	rw	rw

Bits 31:24 Reserved, must be kept at reset value.

Bits 23:20 **L[3:0]**: Regular channel sequence length

These bits are written by software to define the total number of conversions in the regular channel conversion sequence.

0000: 1 conversion

0001: 2 conversions

...

1111: 16 conversions

Bits 19:15 **SQ16[4:0]**: 16th conversion in regular sequence

These bits are written by software with the channel number (0..18) assigned as the 16th in the conversion sequence.

Bits 14:10 **SQ15[4:0]**: 15th conversion in regular sequence

Bits 9:5 **SQ14[4:0]**: 14th conversion in regular sequence

Bits 4:0 **SQ13[4:0]**: 13th conversion in regular sequence

13.13.10 ADC regular sequence register 2 (ADC_SQR2)

Address offset: 0x30

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		SQ12[4:0]					SQ11[4:0]					SQ10[4:1]			
		rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SQ10_0	SQ9[4:0]					SQ8[4:0]					SQ7[4:0]				
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:30 Reserved, must be kept at reset value.

Bits 29:26 **SQ12[4:0]**: 12th conversion in regular sequence

These bits are written by software with the channel number (0..18) assigned as the 12th in the sequence to be converted.

Bits 24:20 **SQ11[4:0]**: 11th conversion in regular sequence

Bits 19:15 **SQ10[4:0]**: 10th conversion in regular sequence

Bits 14:10 **SQ9[4:0]**: 9th conversion in regular sequence

Bits 9:5 **SQ8[4:0]**: 8th conversion in regular sequence

Bits 4:0 **SQ7[4:0]**: 7th conversion in regular sequence

13.13.11 ADC regular sequence register 3 (ADC_SQR3)

Address offset: 0x34

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		SQ6[4:0]					SQ5[4:0]					SQ4[4:1]			
		rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SQ4_0	SQ3[4:0]					SQ2[4:0]					SQ1[4:0]				
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:30 Reserved, must be kept at reset value.

Bits 29:25 **SQ6[4:0]**: 6th conversion in regular sequence

These bits are written by software with the channel number (0..18) assigned as the 6th in the sequence to be converted.

Bits 24:20 **SQ5[4:0]**: 5th conversion in regular sequence

Bits 19:15 **SQ4[4:0]**: 4th conversion in regular sequence

Bits 14:10 **SQ3[4:0]**: 3rd conversion in regular sequence

Bits 9:5 **SQ2[4:0]**: 2nd conversion in regular sequence

Bits 4:0 **SQ1[4:0]**: 1st conversion in regular sequence

13.13.12 ADC injected sequence register (ADC_JSQR)

Address offset: 0x38

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										JL[1:0]		JSQ4[4:1]			
										rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
JSQ4[0]		JSQ3[4:0]				JSQ2[4:0]				JSQ1[4:0]					
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:22 Reserved, must be kept at reset value.

Bits 21:20 **JL[1:0]**: Injected sequence length

These bits are written by software to define the total number of conversions in the injected channel conversion sequence.

00: 1 conversion

01: 2 conversions

10: 3 conversions

11: 4 conversions

Bits 19:15 **JSQ4[4:0]**: 4th conversion in injected sequence (when JL[1:0]=3, see note below)

These bits are written by software with the channel number (0..18) assigned as the 4th in the sequence to be converted.

Bits 14:10 **JSQ3[4:0]**: 3rd conversion in injected sequence (when JL[1:0]=3, see note below)

Bits 9:5 **JSQ2[4:0]**: 2nd conversion in injected sequence (when JL[1:0]=3, see note below)

Bits 4:0 **JSQ1[4:0]**: 1st conversion in injected sequence (when JL[1:0]=3, see note below)

Note: When JL[1:0]=3 (4 injected conversions in the sequencer), the ADC converts the channels in the following order: JSQ1[4:0], JSQ2[4:0], JSQ3[4:0], and JSQ4[4:0].

When JL=2 (3 injected conversions in the sequencer), the ADC converts the channels in the following order: JSQ2[4:0], JSQ3[4:0], and JSQ4[4:0].

When JL=1 (2 injected conversions in the sequencer), the ADC converts the channels in starting from JSQ3[4:0], and then JSQ4[4:0].

When JL=0 (1 injected conversion in the sequencer), the ADC converts only JSQ4[4:0] channel.

13.13.13 ADC injected data register x (ADC_JDRx) (x= 1..4)

Address offset: 0x3C - 0x48

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
JDATA[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **JDATA[15:0]**: Injected data

These bits are read-only. They contain the conversion result from injected channel x. The data are left- or right-aligned as shown in [Figure 48](#) and [Figure 49](#).

13.13.14 ADC regular data register (ADC_DR)

Address offset: 0x4C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **DATA[15:0]**: Regular data

These bits are read-only. They contain the conversion result from the regular channels. The data are left- or right-aligned as shown in [Figure 48](#) and [Figure 49](#).

13.13.15 ADC Common status register (ADC_CSR)

Address offset: 0x00 (this offset address is relative to ADC1 base address + 0x300)

Reset value: 0x0000 0000

This register provides an image of the status bits of the different ADCs. Nevertheless it is read-only and does not allow to clear the different status bits. Instead each status bit must be cleared by writing it to 0 in the corresponding ADC_SR register.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										OVR3	STRT3	JSTRT3	JEOC 3	EOC3	AWD3
										ADC3					
										r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		OVR2	STRT2	JSTRT 2	JEOC2	EOC2	AWD2	Reserved		OVR1	STRT1	JSTRT1	JEOC 1	EOC1	AWD1
		ADC2					ADC1								
		r	r	r	r	r	r			r	r	r	r	r	

Bits 31:22 Reserved, must be kept at reset value.

Bit 21 **OVR3**: Overrun flag of ADC3

This bit is a copy of the OVR bit in the ADC3_SR register.

Bit 20 **STRT3**: Regular channel Start flag of ADC3

This bit is a copy of the STRT bit in the ADC3_SR register.

Bit 19 **JSTRT3**: Injected channel Start flag of ADC3

This bit is a copy of the JSTRT bit in the ADC3_SR register.

Bit 18 **JEOC3**: Injected channel end of conversion of ADC3

This bit is a copy of the JEOC bit in the ADC3_SR register.

Bit 17 **EOC3**: End of conversion of ADC3

This bit is a copy of the EOC bit in the ADC3_SR register.

Bit 16 **AWD3**: Analog watchdog flag of ADC3

This bit is a copy of the AWD bit in the ADC3_SR register.

Bits 15:14 Reserved, must be kept at reset value.

Bit 13 **OVR2**: Overrun flag of ADC2

This bit is a copy of the OVR bit in the ADC2_SR register.

Bit 12 **STRT2**: Regular channel Start flag of ADC2

This bit is a copy of the STRT bit in the ADC2_SR register.

Bit 11 **JSTRT2**: Injected channel Start flag of ADC2

This bit is a copy of the JSTRT bit in the ADC2_SR register.

Bit 10 **JEOC2**: Injected channel end of conversion of ADC2

This bit is a copy of the JEOC bit in the ADC2_SR register.

Bit 9 **EOC2**: End of conversion of ADC2

This bit is a copy of the EOC bit in the ADC2_SR register.

Bit 8 **AWD2**: Analog watchdog flag of ADC2

This bit is a copy of the AWD bit in the ADC2_SR register.

Bits 7:6 Reserved, must be kept at reset value.

Bit 5 **OVR1**: Overrun flag of ADC1

This bit is a copy of the OVR bit in the ADC1_SR register.

Bit 4 **STRT1**: Regular channel Start flag of ADC1

This bit is a copy of the STRT bit in the ADC1_SR register.

Bit 3 **JSTRT1**: Injected channel Start flag of ADC1

This bit is a copy of the JSTRT bit in the ADC1_SR register.

Bit 2 **JEOC1**: Injected channel end of conversion of ADC1

This bit is a copy of the JEOC bit in the ADC1_SR register.

Bit 1 **EOC1**: End of conversion of ADC1

This bit is a copy of the EOC bit in the ADC1_SR register.

Bit 0 **AWD1**: Analog watchdog flag of ADC1

This bit is a copy of the AWD bit in the ADC1_SR register.

13.13.16 ADC common control register (ADC_CCR)

Address offset: 0x04 (this offset address is relative to ADC1 base address + 0x300)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								TSVREFE	VBATE	Reserved				ADCPRE	
								rw	rw					rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DMA[1:0]		DDS	Res.	DELAY[3:0]				Reserved			MULTI[4:0]				
rw	rw	rw		rw	rw	rw	rw				rw	rw	rw	rw	rw

Bits 31:24 Reserved, must be kept at reset value.

Bit 23 **TSVREFE**: Temperature sensor and V_{REFINT} enable

This bit is set and cleared by software to enable/disable the temperature sensor and the V_{REFINT} channel.

0: Temperature sensor and V_{REFINT} channel disabled

1: Temperature sensor and V_{REFINT} channel enabled

Note: On STM32F42x and STM32F43x devices, VBATE must be disabled when TSVREFE is set. If both bits are set, only the VBAT conversion is performed.

Bit 22 **VBATE**: V_{BAT} enable

This bit is set and cleared by software to enable/disable the V_{BAT} channel.

0: V_{BAT} channel disabled

1: V_{BAT} channel enabled

Bits 21:18 Reserved, must be kept at reset value.

Bits 17:16 **ADCPRE**: ADC prescaler

Set and cleared by software to select the frequency of the clock to the ADC. The clock is common for all the ADCs.

Note: 00: PCLK2 divided by 2
01: PCLK2 divided by 4
10: PCLK2 divided by 6
11: PCLK2 divided by 8

Bits 15:14 **DMA**: Direct memory access mode for multi ADC mode

This bit-field is set and cleared by software. Refer to the DMA controller section for more details.

00: DMA mode disabled
01: DMA mode 1 enabled (2 / 3 half-words one by one - 1 then 2 then 3)
10: DMA mode 2 enabled (2 / 3 half-words by pairs - 2&1 then 1&3 then 3&2)
11: DMA mode 3 enabled (2 / 3 bytes by pairs - 2&1 then 1&3 then 3&2)

Bit 13 **DDS**: DMA disable selection (for multi-ADC mode)

This bit is set and cleared by software.

0: No new DMA request is issued after the last transfer (as configured in the DMA controller). DMA bits are not cleared by hardware, however they must have been cleared and set to the wanted mode by software before new DMA requests can be generated.

1: DMA requests are issued as long as data are converted and DMA = 01, 10 or 11.

Bit 12 Reserved, must be kept at reset value.

Bit 11:8 **DELAY**: Delay between 2 sampling phases

Set and cleared by software. These bits are used in dual or triple interleaved modes.

0000: $5 * T_{ADCCLK}$

0001: $6 * T_{ADCCLK}$

0010: $7 * T_{ADCCLK}$

...

1111: $20 * T_{ADCCLK}$

Bits 7:5 Reserved, must be kept at reset value.

Bits 4:0 **MULTI[4:0]**: Multi ADC mode selection

These bits are written by software to select the operating mode.

- All the ADCs independent:
00000: Independent mode
- 00001 to 01001: Dual mode, ADC1 and ADC2 working together, ADC3 is independent
00001: Combined regular simultaneous + injected simultaneous mode
00010: Combined regular simultaneous + alternate trigger mode
00011: Reserved
00101: Injected simultaneous mode only
00110: Regular simultaneous mode only
00111: interleaved mode only
01001: Alternate trigger mode only
- 10001 to 11001: Triple mode: ADC1, 2 and 3 working together
10001: Combined regular simultaneous + injected simultaneous mode
10010: Combined regular simultaneous + alternate trigger mode
10011: Reserved
10101: Injected simultaneous mode only
10110: Regular simultaneous mode only
10111: interleaved mode only
11001: Alternate trigger mode only

All other combinations are reserved and must not be programmed

Note: In multi mode, a change of channel configuration generates an abort that can cause a loss of synchronization. It is recommended to disable the multi ADC mode before any configuration change.

13.13.17 ADC common regular data register for dual and triple modes (ADC_CDR)

Address offset: 0x08 (this offset address is relative to ADC1 base address + 0x300)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DATA2[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA1[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:16 **DATA2[15:0]**: 2nd data item of a pair of regular conversions

- In dual mode, these bits contain the regular data of ADC2. Refer to [Dual ADC mode](#).
- In triple mode, these bits contain alternatively the regular data of ADC2, ADC1 and ADC3. Refer to [Triple ADC mode](#).

Bits 15:0 **DATA1[15:0]**: 1st data item of a pair of regular conversions

- In dual mode, these bits contain the regular data of ADC1. Refer to [Dual ADC mode](#).
- In triple mode, these bits contain alternatively the regular data of ADC1, ADC3 and ADC2. Refer to [Triple ADC mode](#).

13.13.18 ADC register map

The following table summarizes the ADC registers.

Table 71. ADC global register map

Offset	Register
0x000 - 0x04C	ADC1
0x050 - 0x0FC	Reserved
0x100 - 0x14C	ADC2
0x118 - 0x1FC	Reserved
0x200 - 0x24C	ADC3
0x250 - 0x2FC	Reserved
0x300 - 0x308	Common registers

Table 72. ADC register map and reset values for each ADC

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0				
0x00	ADC_SR	Reserved																										OVR	STRT	JSTRT	JEOC	EOC	AWD				
	Reset value																											0	0	0	0	0	0				
0x04	ADC_CR1	Reserved					OVRIE	RES[1:0]		AWDEN	JAWDEN	Reserved				DISC NUM [2:0]			JDISCEN	DISCEN	AUTO	AWD SGL	SCAN	JEOCIE	AWDIE	EOCIE	AWDCH[4:0]										
	Reset value						0	0	0	0	0					0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				
0x08	ADC_CR2	Reset value	SWSTART	EXTEN[1:0]		EXTSEL [3:0]				Reset value	JSWSTART	JEXTEN[1:0]		JEXTSEL [3:0]			Reserved			ALIGN	EOCS	DDS	DMA	Reserved					CONT	ADON							
	Reset value		0	0	0	0	0	0	0	0		0	0	0	0	0	0	0	0	0	0	0	0						0	0							
0x0C	ADC_SMPR1	Sample time bits SMPx_x																																			
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				
0x10	ADC_SMPR2	Sample time bits SMPx_x																																			
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				
0x14	ADC_JOFR1	Reserved																		JOFFSET1[11:0]																	
	Reset value																			0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x18	ADC_JOFR2	Reserved																		JOFFSET2[11:0]																	
	Reset value																			0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x1C	ADC_JOFR3	Reserved																		JOFFSET3[11:0]																	
	Reset value																			0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x20	ADC_JOFR4	Reserved																		JOFFSET4[11:0]																	
	Reset value																			0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x24	ADC_HTR	Reserved																		HT[11:0]																	
	Reset value																			1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0x28	ADC_LTR	Reserved																		LT[11:0]																	
	Reset value																			0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x2C	ADC_SQR1	Reserved								L[3:0]		Regular channel sequence SQx_x bits																									
	Reset value									0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
0x30	ADC_SQR2	Reserved	Regular channel sequence SQx_x bits																																		
	Reset value		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					
0x34	ADC_SQR3	Reserved	Regular channel sequence SQx_x bits																																		
	Reset value		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					
0x38	ADC_JSQR	Reserved								JL[1:0]		Injected channel sequence JSQx_x bits																									
	Reset value									0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
0x3C	ADC_JDR1	Reserved														JDATA[15:0]																					
	Reset value															0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x40	ADC_JDR2	Reserved														JDATA[15:0]																					
	Reset value															0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x44	ADC_JDR3	Reserved														JDATA[15:0]																					
	Reset value															0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x48	ADC_JDR4	Reserved														JDATA[15:0]																					
	Reset value															0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x4C	ADC_DR	Reserved														Regular DATA[15:0]																					
	Reset value															0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 73. ADC register map and reset values (common ADC registers)

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0										
0x00	ADC_CSR	Reserved											OVR	STRT	JSTRT	JEOC	EOC	AWD	Reserved	Reserved	OVR	STRT	JSTRT	JEOC	EOC	AWD	Reserved	Reserved	OVR	STRT	JSTRT	JEOC	EOC	AWD									
	Reset value												0	0	0	0	0	0			0	0	0	0	0	0			0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
													ADC3						ADC2						ADC1																		
0x04	ADC_CCR	Reserved									TSVREFE	VBATE	Reserved			DMA[1:0]		DDS		Reserved	DELAY [3:0]			Reserved			MULTI [4:0]																
	Reset value										0	0																			0	0	0	0	0	0	0	0	0	0	0	0	0
0x08	ADC_CDR	Regular DATA2[15:0]																													Regular DATA1[15:0]												
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0									

Refer to [Section 2.3: Memory map](#) for the register boundary addresses.